

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **SADHANA S (192524298)**_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SADHANA S

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Backtracking Search

Given Constraints

- D1 cannot work **Night**
- D2 must work before D3 (**Morning** < **Afternoon** < **Night**)
- One doctor per shift
- D3 cannot work **Morning**
- Every doctor gets exactly one shift

Backtracking Steps

1. Assign **D1** → **Morning**
2. Assign **D2** → **Afternoon**
3. Assign **D3** → **Night**

Constraint Check

- D1 not in Night
- D2 before D3
- One doctor per shift
- D3 not in Morning
- All doctors assigned

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Valid schedule found using Backtracking Search.

2. BFS Search Algorithm

Constraints:

- Grid size = 5×5
- Movement = Up, Down, Left, Right
- Cost per move = 1
- Obstacles cannot be crossed
- Manhattan Distance heuristic: $|x_1 - x_2| + |y_1 - y_2|$

BFS Node Expansion

Step	Expanded Node	Queue
1	S	Adjacent nodes of S
2	First adjacent node	Remaining nodes + new neighbors
3	Next node	Updated queue
4	Continue level by level	Updated queue
5	G	Goal Reached

Optimal Path

BFS explores nodes **level by level** and returns the **shortest path** (minimum number of moves) from **S** to **G**, while avoiding obstacles.

Total Cost

$$\boxed{\text{Cost} = \text{Number of moves} \times 1}$$

Result

Since the grid and obstacle positions are not provided, the **exact path and total cost cannot be determined**. However, **BFS guarantees the shortest path** when each move has the same cost (1).

3. Autonomous Rescue Robot

(a) Constraint Analysis

- Robot moves only Up, Down, Left, Right.
- Cannot cross obstacles.
- Limited sensing (adjacent cells only).
- Each move costs 1.
- Risky zones add cost +2.
- Robot updates knowledge while moving.

These constraints help the robot choose the safest and lowest-cost path.

(b) Solution Approach (Online Search / Uniform Cost Search)

1. Start from S.
2. Observe nearby cells.
3. Avoid blocked paths.
4. Calculate movement cost.
5. Expand the lowest-cost path.
6. Update map when new information is found.
7. Continue until Goal (G) is reached.

(c) AI Concepts

Intelligent Agent

- Perceives the environment and takes actions.

Rationality

- Chooses the safest and minimum-cost path.

Environment Type

- Partially observable
- Dynamic
- Sequential
- Discrete

(d) Justification

Uniform Cost Search (or Online Search Agent) is suitable because:

- Finds the minimum-cost path.
- Avoids risky and blocked areas.

- Works well with changing environments.
- Updates decisions using newly observed information.

Conclusion:

The robot safely reaches the survivor while minimizing total travel cost.